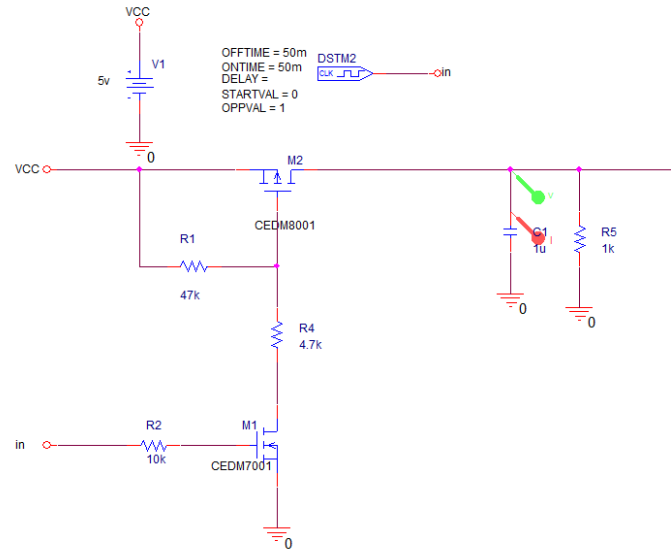


HW 15

엄민호

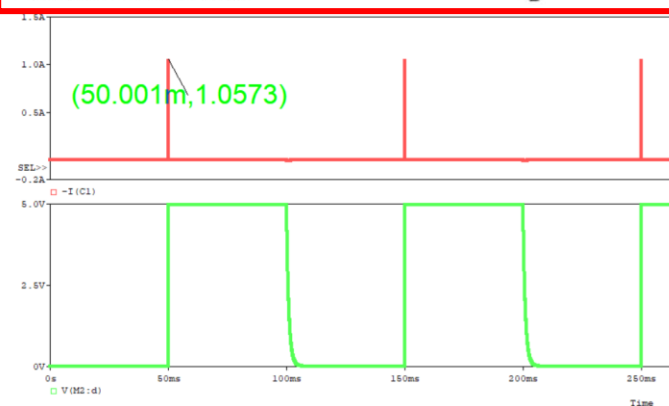
Load switch - inrush current problem



Load의 허용 순간최대 전류 : 500mA
CEDM8001(P-channel MOS) Continuous I_d
MAX : 200mA

MAXIMUM RATINGS: ($T_A=25^\circ\text{C}$)

	SYMBOL		UNITS
Drain-Source Voltage	V_{DS}	20	V
Gate-Source Voltage	V_{GS}	10	V
Continuous Drain Current (Steady State)	I_D	100	mA
Continuous Drain Current	I_D	200	mA



실제 load에는 Capacitance 존재

M2 MOSFET이 Turn-on 될 때 capacitor 에 inrush current 발생 (1.05A)

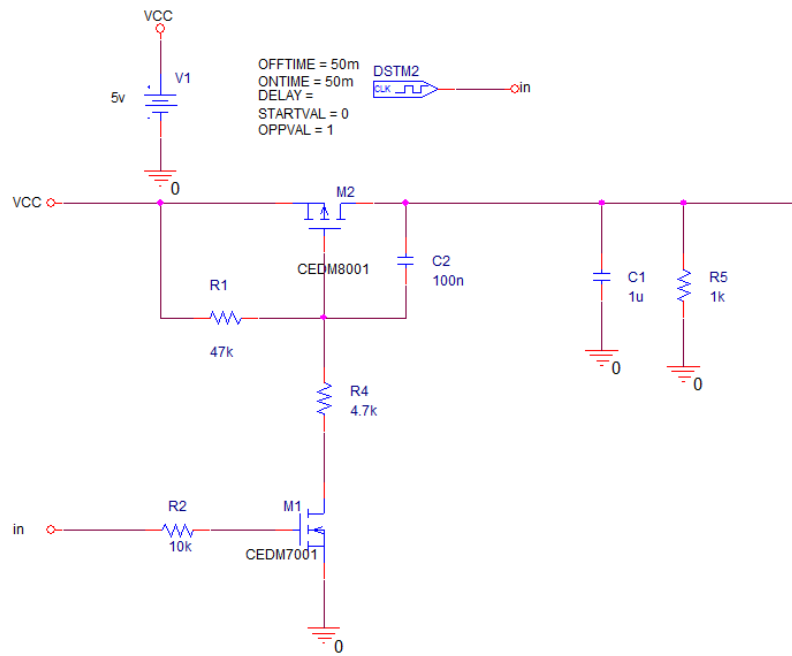
Turn-off 될 때 capacitor가 방전되며 지연 발생(고속회로 아니면 큰 문제 x)

Load의 허용 순간최대 전류를 초과 -> **대책필요**

MOSFET datasheet에 허용순간최대 전류 명시X -> Continuous I_d MAX 를 넘지 않도록 설계 선택 -

> **Inrush current 200mA 이하 설계**

Inrush Current Protection

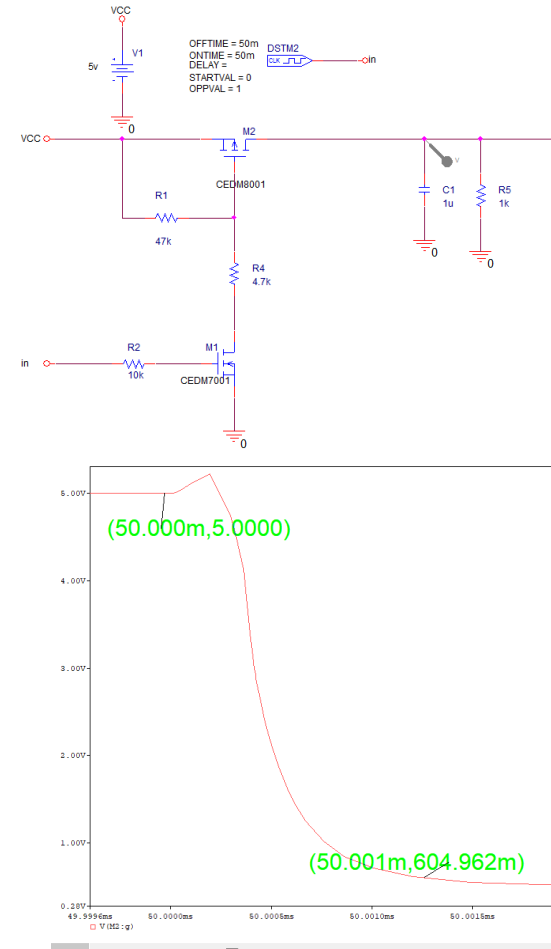
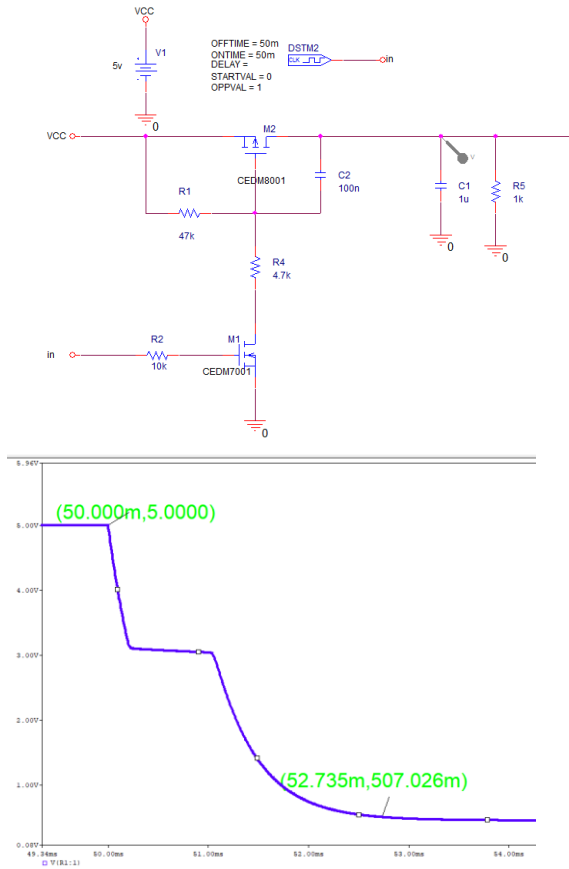


Gate - Drain 사이에 캐패시터(C2- $\rightarrow C_{gs}$)를 연결 시 예상 동작 :

- ➔ M1 MOS turn-on 때 M2 MOS의 Gate 전압 부드럽게 떨어짐
- ➔ M2 MOS P-channel 형성에 지연
- ➔ 스위칭 속도 지연
- ➔ Capacitor의 Voltage rising time 증가
- ➔ Inrush current 감소

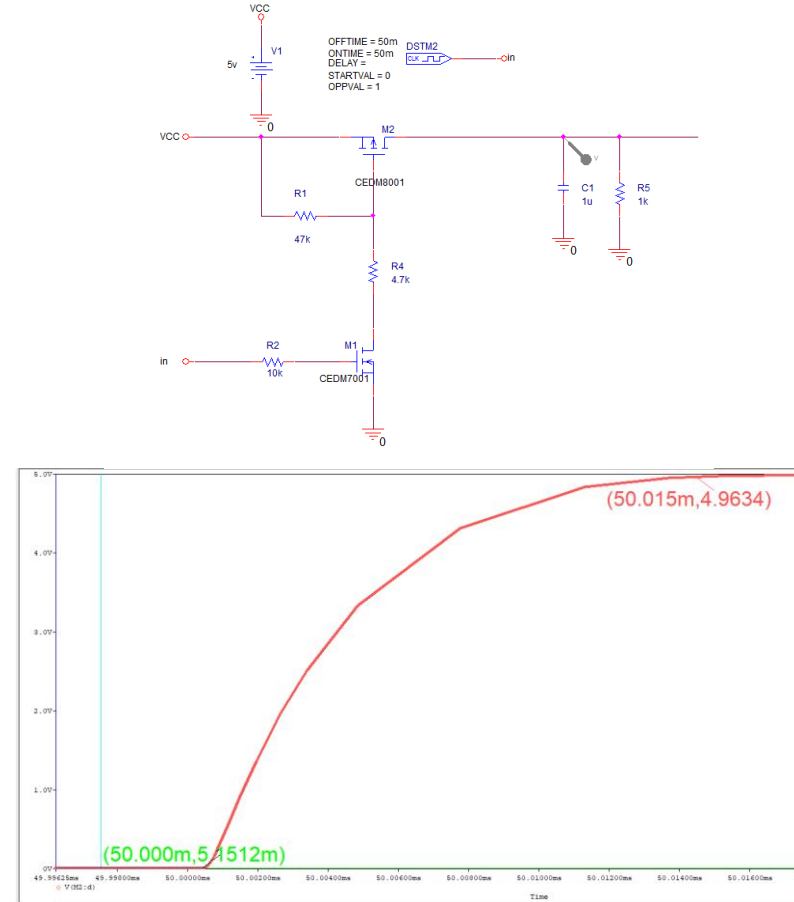
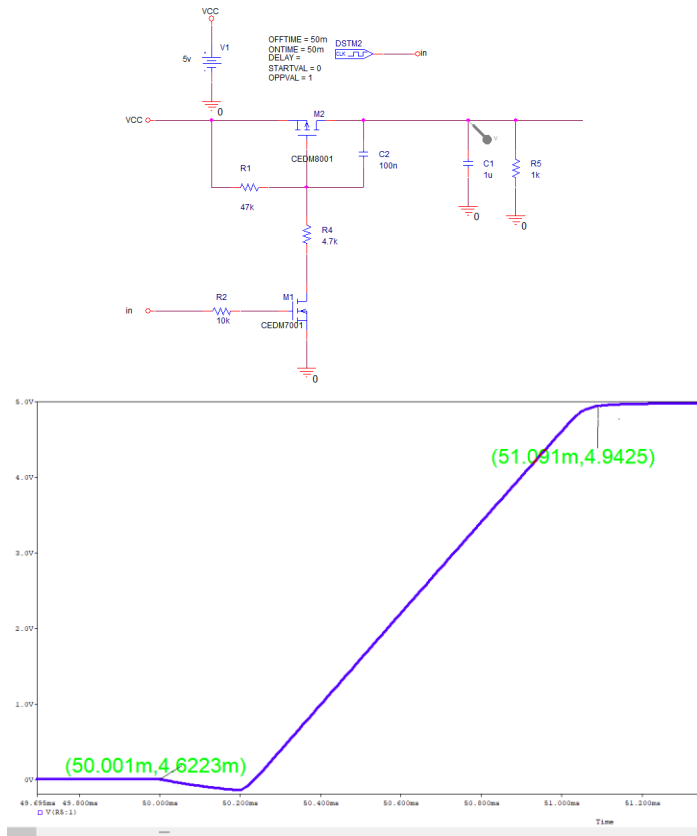
C2의 Capacitance 에 따른 동작변화 관측 필요

Gate 입력전압 변화 비교



C_{gs} 추가 시 Gate 전압 부드럽고 느리게 떨어짐

Load capacitor에 걸리는 전압 비교

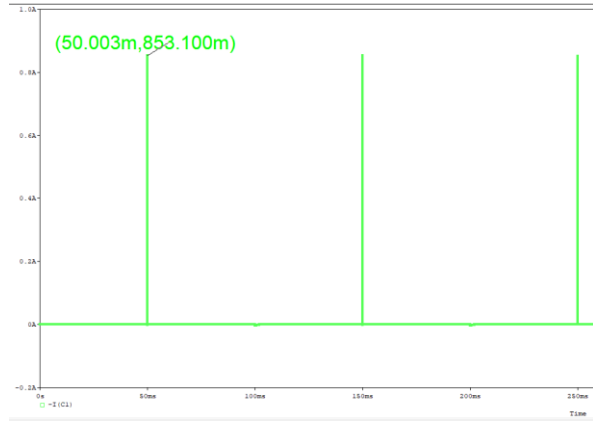


C_{gs} 추가시 C1 capacitor에 걸리는 voltage rising time이 길어짐
(약 0.015ms -> 1ms)

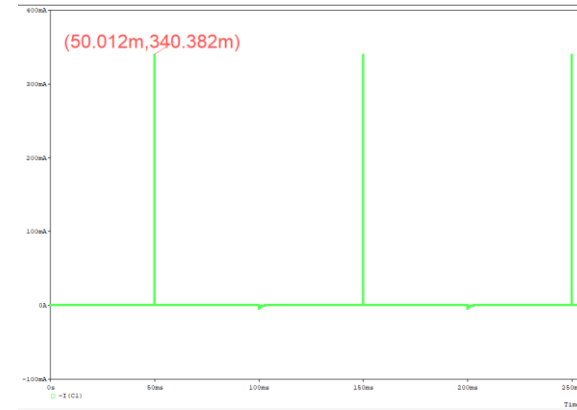
$$I_c = C \times \frac{dv_c}{dt}$$

-> Inrush current 감소

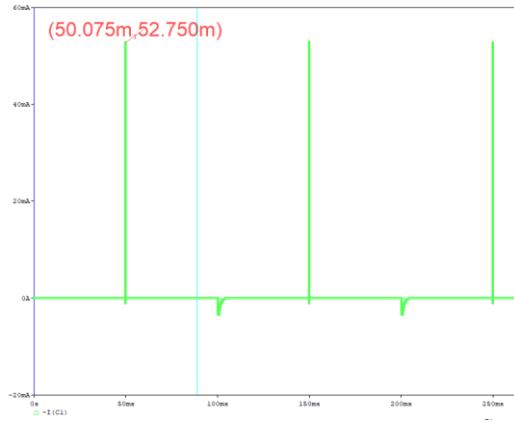
a. C_{gs} 의 Capacitance 크기에 따른 Inrush current 크기 비교 (100pF/1nF/10nF/100nF)



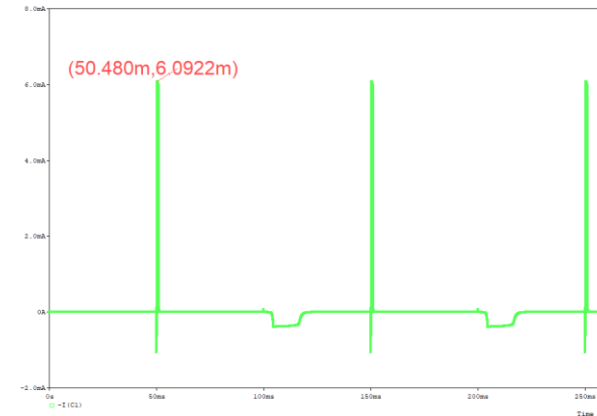
100pF > → 853mA



1nF > → 340mA



10nF > → 53.8mA



100nF > → 6.1mA

b. C_{gs} 의 Capacitance 크기 선정 (100pF/1nF/10nF/100nF)

1. CEDM8001(P-channel MOS) Continuous I_d MAX : 200mA

-> Inrush current 의 크기 200mA 넘지 않도록

(10nF > $\rightarrow$ 53.8mA or 100nF > $\rightarrow$ 6.1mA)

2. Capacitance $\uparrow$ - Gate 전압의 변화가 느림 -> dynamic state의 시간이 길어짐 -> MOSFET 소모 전력 증가

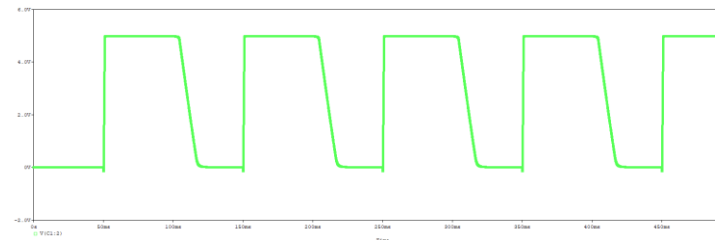
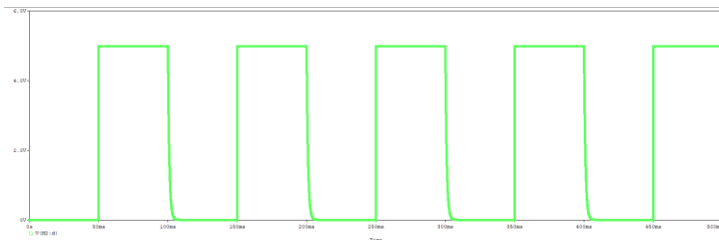
(10nF better)

3. Capacitance $\uparrow$ -> 출력전압의 스위칭 속도가 느려짐 -> 고속신호 동작 어려움

Vout에서의 Rising time 지연도 길어지지만 falling time은 더 길어짐 (Capacitor 방전시간)

방전시간 $\uparrow$ -> 전력손실 $\uparrow$

(10nF better)

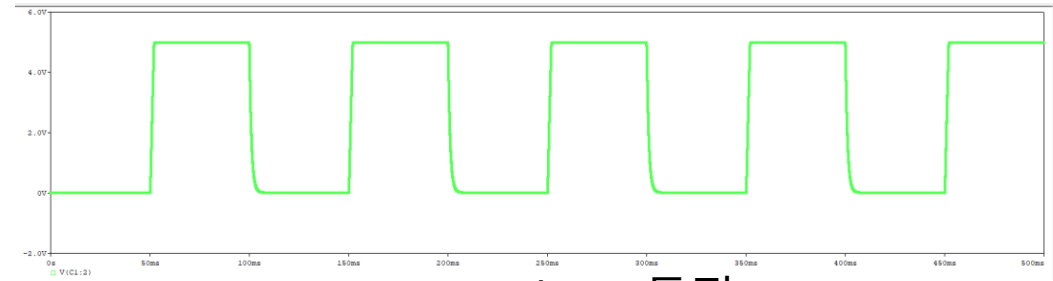
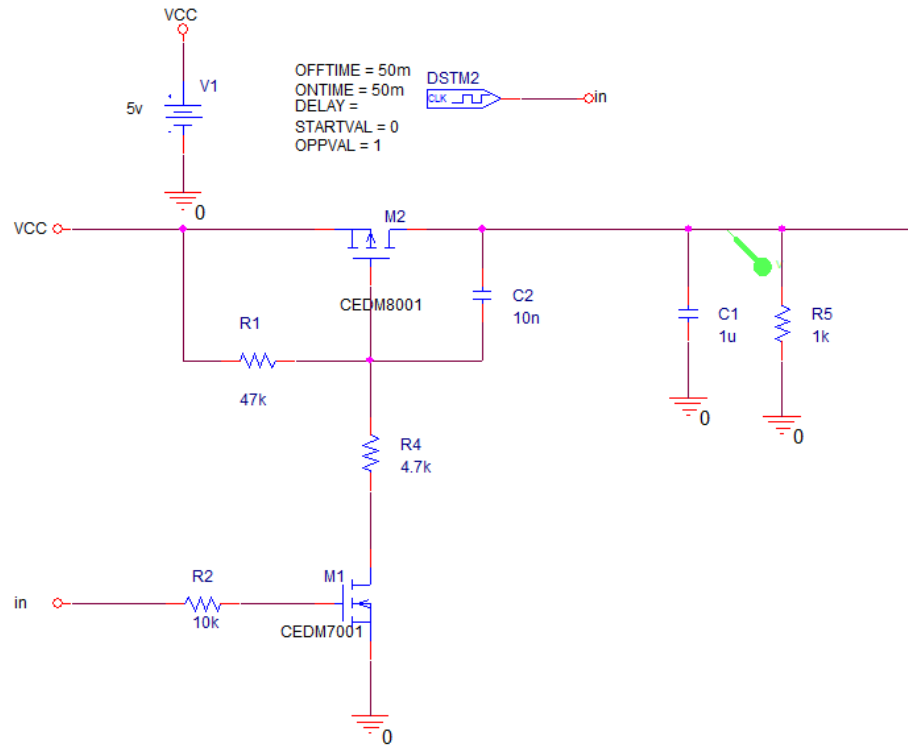


Capacitance $\uparrow$ - inrush current $\downarrow$, 전력손실 $\uparrow$, 고속신호 동작 x
Capacitance $\downarrow$ - inrush current $\uparrow$, 전력손실 $\uparrow$, 고속신호 동작 o

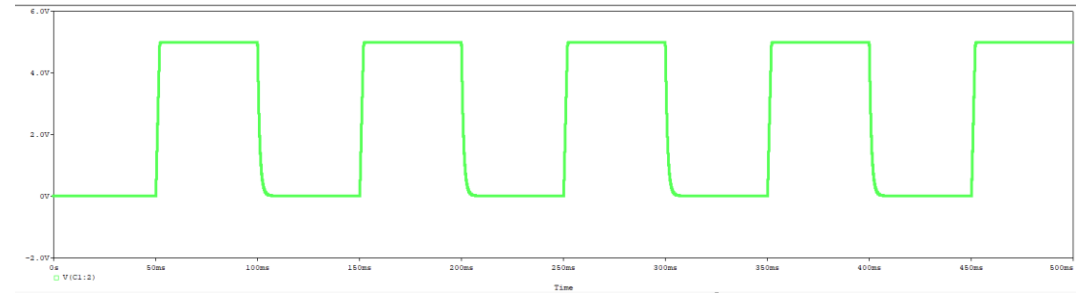


10nF Cap 선정

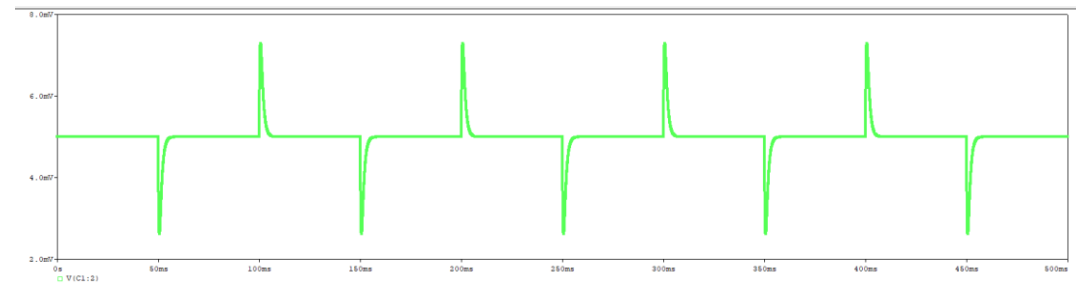
c. Gate 저항(R_g, R_4) 에 따른 출력전압 ($4.7k\Omega$ $47k\Omega$, $470k\Omega$)



<4.7k Ω > 동작o

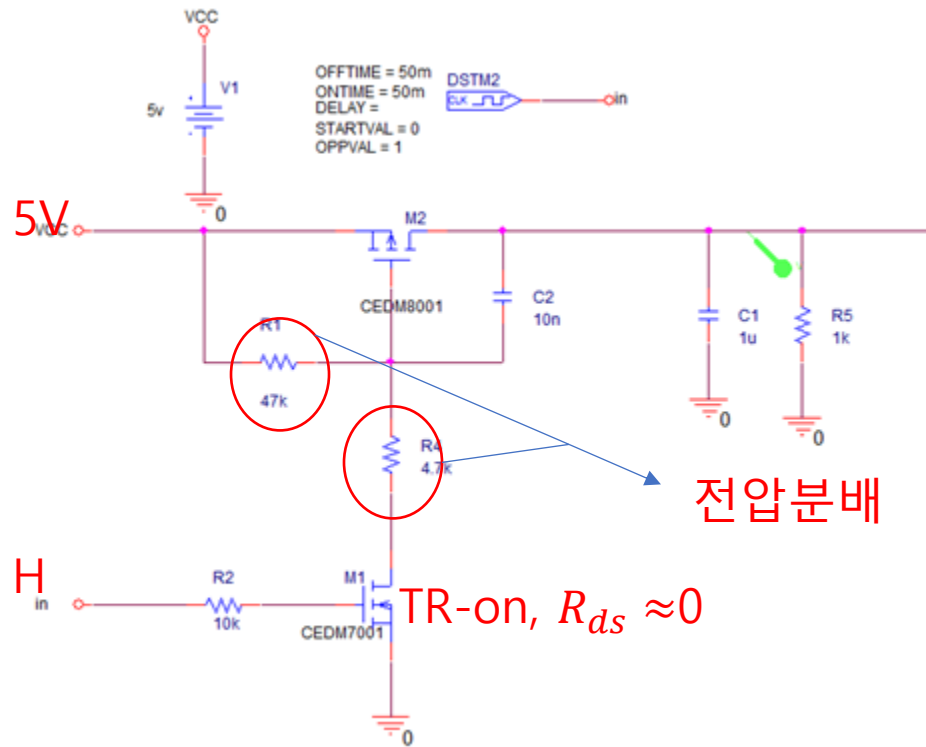


<47k Ω > 동작o



<470k Ω > 동작x

c. Gate 저항(R_g, R_4) 에 따른 출력전압 (4.7k Ω 47k Ω , 470k Ω)



전압분배

TR-on, $R_{ds} \approx 0$

ELECTRICAL CHARACTERISTICS: ($T_A=25^\circ\text{C}$ unless otherwise noted)					
SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
I_{GSSF}, I_{GSSR}	$V_{GS}=10\text{V}, V_{DS}=0$			1.0	μA
I_{DSS}	$V_{DS}=20\text{V}, V_{GS}=0$			1.0	μA
BV_{DSS}	$V_{GS}=0, I_D=100\mu\text{A}$	20			V
$V_{GS(th)}$	$V_{DS}=V_{GS}, I_D=250\mu\text{A}$	0.6		1.1	V
$r_{DS(ON)}$	$V_{GS}=4.0\text{V}, I_D=10\text{mA}$		1.9	8.0	Ω
$r_{DS(ON)}$	$V_{GS}=2.5\text{V}, I_D=10\text{mA}$		2.4	12	Ω
$r_{DS(ON)}$	$V_{GS}=1.5\text{V}, I_D=1.0\text{mA}$			45	Ω
θ_{FS}	$V_{DS}=10\text{V}, I_D=100\text{mA}$	100			mS
C_{rss}	$V_{DS}=3.0\text{V}, V_{GS}=0, f=1.0\text{MHz}$		15		pF
C_{iss}	$V_{DS}=3.0\text{V}, V_{GS}=0, f=1.0\text{MHz}$		45		pF
C_{oss}	$V_{DS}=3.0\text{V}, V_{GS}=0, f=1.0\text{MHz}$		15		pF
$Q_{g(tot)}$	$V_{DS}=10\text{V}, V_{GS}=4.5\text{V}, I_D=100\text{mA}$		0.658		nC
Q_{gs}	$V_{DS}=10\text{V}, V_{GS}=4.5\text{V}, I_D=100\text{mA}$		0.158		nC
Q_{gd}	$V_{DS}=10\text{V}, V_{GS}=4.5\text{V}, I_D=100\text{mA}$		0.181		nC
t_{on}	$V_{DD}=3.0\text{V}, V_{GS}=2.5\text{V}, I_D=10\text{mA}$		35		ns
t_{off}	$V_{DD}=3.0\text{V}, V_{GS}=2.5\text{V}, I_D=10\text{mA}$		80		ns

R4가 없으면 MOSFET이 노이즈에 취약 및 inrush current 동작 시 수명 감소.
R4가 커지면서 소비전력 증가
R4가 너무 크면 TR-on 시에 충분한 $V_{gs(th)}$ 가 생기지 않아서 정상작동 x